CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 06

By- Aditya sir

Recap of Previous Lecture



Topic — Asymptotic Notations

Topic — Practice Questions.

Topics to be Covered



- Topic
- Topic
- Topic

Properties

Questions.



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

[MCQ] P40

#Q. Consider the following functions from positive integers to real number:

$10, \sqrt{n}, n\log_2 n, \frac{100}{n}$

The correct arrangement of the above functions in increasing order of asymptotic complexity is:

A $\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$

B $10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$

C $\frac{100}{n}, 10, \log_2 n\sqrt{n}, n$

D $\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$

$\sqrt{n} < n$

$f_1 = 10 \longrightarrow$ Const

$f_2 = \sqrt{n} \longrightarrow$ Poly

$f_3 = \log_2 n \longrightarrow$ log

$f_4 = n \longrightarrow$ Poly

$f_5 = \frac{100}{n} \longrightarrow$ Decr

$[\text{Decr} < \text{Const} < \log < \text{Poly} < \text{Expo}]$

$\left[\frac{100}{n} < 10 < \log n < \sqrt{n} < n\right]$

$\frac{100}{n} \longrightarrow$ decreasing



Topic : General Properties of Big Oh Notation

Let d(n), e(n), f(n), and g(n) be functions mapping nonnegative integers to nonnegative reals. Then

1. If d(n) is O(f(n)), then ad(n) is O(f(n)), for any constant a > 0
2. If d(n) is O(f(n)) and e(n) is O(g(n)), then d(n) + e(n) is O(f(n) + g(n)).
3. If d(n) is O(f(n)) and e(n) is O(g(n)), then d(n)e(n) is O(f(n)g(n))
4. If d(n) is O(f(n)) and f(n) is O(g(n)), then d(n) is O(g(n)).
5. If f(n) is a polynomial of degree d (that is, $f(n) = (a_0 + a_1 n + ... + a_d n^d)$ then f(n) is ~~O(nd)~~ $O(n^d)$.
6. n^x is O(a") for any fixed x > 0 and a > 1
7. $\log(n^x)$ is O(log n) for any fixed x > 0
8. $\log^x n$ is $O(n^y)$ for any fixed constants x > 0 and y > 0



Topic : General Properties of Big Oh Notation

1. If $d(n) = O(f(n))$,

   then $a * d(n) = O(f(n))$, $a > 0$

e.g.: $d(n) = n^2$ → $O(n^2)$

$a = 10$

$a * d(n) = 10 * n^2$ → $O(n^2)$

$O(10 * n^2) = O(n^2)$

$O(C * F(n)) = O(f(n))$



Topic : General Properties of Big Oh Notation

2. $d(n) = 5n + 2 \rightarrow O(n)$

$e(n) = 10n^2 + 7n + 9 \rightarrow O(n^2)$

$d(n) + e(n) = (10n^2 + 7n + 9) + (5n + 2)$ $\Rightarrow O(f(n) + g(n))$

$= (10n^2 + 12n + 11)$ $\Rightarrow O(n + n^2)$

$= O(n^2)$ $\Rightarrow O(n^2)$

Shortcut: $O(f(n) + g(n)) = O(\max(f(n), g(n)))$

$\max(n, n^2) = n^2$



Topic : General Properties of Big Oh Notation

3. $d(n) = 5n \rightarrow O(n)$

$e(n) = 10n^2 + 2 \rightarrow O(n^2)$

$d(n) * e(n) = (10n^2 + 2) * (5n)$

$= (50n^3 + 10n)$

$= O(n^3)$

$\Rightarrow O(f(n) * g(n))$

$\Rightarrow O(n * n^2)$

$\Rightarrow O(n^3)$



Topic : General Properties of Big Oh Notation

4. $d(n) = O(f(n))$ and $f(n) = O(g(n)) \Rightarrow d(n) = O(g(n))$

e.g.:

$d(n) = O(n^2)$

and $f(n) = O(n^4)$

then $d(n) = O(n^4)$

$d \to O(n^2)$

$n^2 = O(n^4)$

$d \to O(n^4)$

$a \leq b$

$b \leq c$

$a \leq c$



Topic : General Properties of Big Oh Notation

5. $f(n) = 5 + 10n + 15n^3 + 7n^4$

$f(n) = O(n^4)$



Topic : General Properties of Big Oh Notation

6. $n^x = O(a^n)$, for any $x > 0$ & $a > 1$

$n^x, x > 0 \rightarrow$ Polynomial

$a^n, a > 1 \rightarrow$ Exponential

Poly < Expo

Poly = O(Expo)

$n^2 = O(2^n)$?



Topic : General Properties of Big Oh Notation

7. $f(n) = \log(n^x) = x * \log(n) = O(\log n)$

$f(n) = \log(n^3) = 3 * \log(n)$

$f(n) = \log(n^6) = 6 * \log(n)$

$f(n) = \log(n^{10}) = 10 * \log(n)$

$f(n) = \log(n^{100}) = 100 * \log(n)$



Topic : General Properties of Big Oh Notation

8. $(\log n)^x$ or $\log^x(n) = O(n^y)$, $x > 0, y > 0$

Logarithmic < Polynomial

log = O(poly)

$(\log n)^{10} \Rightarrow \log^{10}(n)$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(1) If $0 < x < y$, then $n^x = O(n^y)$

T

$0 < 2 < 4$

$n^2 < n^4$

$n^2 = O(n^4)$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(2) log (n) is $\Omega(1/n)$ (T)

$\log n \rightarrow \log$ ↑

$1/n \rightarrow$ Decr ↓

$\log n > 1/n$

$\Omega(1/n)$ ✓



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(3) 2^{n^2} is O(n!)

$n! \approx n^n$

$2^{n^2} > n! \Rightarrow 2^{n^2} \neq O(n!)$

$n > \log n$

2^{n^2} $\quad$ n^n $\quad$ (F)

Take $\log_2(\,)$

$\log_2(2^{n^2})$ $\quad$ $\log(n^n)$

n^2 $\quad$ $n\log n$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(4) 20*n*log n = O(n log n)

(T)

$a \times f(n) = O(f(n))$

$O(20\ n\log n) = O(n\log n)$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(5) $(n+c)^k \neq O(n^k)$ for some $k > 0, c > 0$ (F)

eg $(n+2)^2$

$= n^2+4n+4 = O(n^2)$

$(n+c)^k \rightarrow$ Poly of degree k.

$= O(n^k)$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(6) n^2 is $O(2^{(\log n)})$

F

$n^2 > 2^{\log n}$

$\log_2(\ )$

$\log_2(n^2)$ $\quad$ $\log(2^{\log n})$

$2 * \log n$ $\quad$ $\log n * \log_2 2$

$2\log n > \log n * 1$

$\log n + \log n$ $\quad$ $\log n$

$\log n > 1$



Topic : Adding Functions

The sum of two functions is governed by the dominant one, namely:

$O(f(n)) + O(g(n)) \rightarrow 0\ (\max(f(n), g(n)))$

$\Omega f(n)) + \Omega(g(n)) \rightarrow \Omega(\max(f(n), g(n)))$

$\theta(f(n)) + \theta(g(n)) \rightarrow \theta(\max(f(n), g(n)))$

$4n^2 + 2n \rightarrow \max(n, n^2)$

$= \underline{\Omega(n^2)}$

$f(n) = 2n \longrightarrow \Omega(n)$

$g(n) = 4n^2 \longrightarrow \Omega(n^2)$



Topic : Adding Functions

Example:

$f(n) = 5n^2 + 2 \quad \rightarrow \quad O(n^2)$

$g(n) = 10n^3 \quad \rightarrow \quad O(n^3)$

$O(f(n)) = n^2$ and $O(g(n)) = n^3$

$n^2 + n^3 = O(\max(5n^2 + 2, 10n^3))$

$= O(10n^3)$

$= O(n^3)$



Topic : Adding Functions

$O(f(n)) * O(g(n)) \rightarrow O(f(n) * g(n))$

$\Omega(f(n)) * \Omega(g(n)) \rightarrow \Omega(f(n) * g(n))$

$\theta(f(n)) * \theta(g(n)) \rightarrow \theta(f(n) * g(n))$



Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

V·V·Imp

(1) $n^2 = O(2^{(2 \log n)})$ (T)

n^2 $2^{(2\log n)}$

$\log_2(\ )$

$\log_2(n^2)$ $\log(2^{2\log n})$

$2\log n$ $2\log n * \log_2 2$

$2\log n = 2\log n$

$2^{2\log n} = \theta(n^2)$

$n^2 = \theta(2^{2\log n})$



Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

(2) $(\log n)^{1/2} = O(\log(\log n))$

F

$>$

Take $\log_2(\ )$

$\log((\log n)^{1/2})$

$\frac{1}{2} * \log(\log n)$

$\log(\log(\log n))$

$\log(\log(\log n)) = x$

$\frac{1}{2} * x > \log(x)$

$\log n = x$

$x^{1/2} = O(\log n)$

$\sqrt{n} > \log n$



Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

(3) $a^n \neq O(n^x)$, for $a > 1, x > 0$

Expo ≠ Pdy

T

$2^n \neq O(n^2)$

$2^n \not\leq n^2$



Topic : Trichotomy Property

#Q. Does Asymptotic notations follow Trichotomy Property?

V. Imp



Topic : Trichotomy Property

Trichotomy Property in Real Numbers:

- Given any two real numbers x & y (fixed).

  Then, x and y follow exactly one of the below relation:

  1) $x = y$ or,

  2) $x > y$ or,

  3) $x < y$

$x < y$

2 5



Topic : Trichotomy Property

Trichotomy Property in Asymptotic Comparison of two function:

- Given two functions f(n) & g(n), does f(n) & g(n) always follow exactly one of the below?

1) $f(n) >_A g(n) \quad \Rightarrow \quad F = \Omega(g), \omega(g)$

or, 2) $f(n) =_A g(n) \quad \Rightarrow \quad F = \theta(g)$

or, 3) $f(n) <_A g(n) \quad \Rightarrow \quad F = O(g), o(g)$



Topic : Trichotomy Property

Example1:

$f(n) = 5n^2$

$g(n) = 7n^3$

$f \overset{A}{<} g$ always

$f = O(g)$,



Topic : Trichotomy Property

Example2:

$f(n) = 5n^2$

$g(n) = 1/n$

$f \gtrsim g$

$\Rightarrow f = \Omega(g)$



Topic : Trichotomy Property

Example3:

$f(n) = 10n^3 + 7$ $\rightarrow \Theta(n^3)$

$g(n) = 5n^3 + 15$ $\rightarrow \Theta(n^3)$

$f \overline{=}_A g$

$f = \Theta(g)$

Topic : Trichotomy Property

Example4: P49

$f(n) = n$

$g(n) = n^{(1+\sin x)}$

Case1 : $\sin x = 1$

$f \to n$

$g \to n^{(1+1)} = n^2$

$f \leq_A g$

$\sin x = -1$

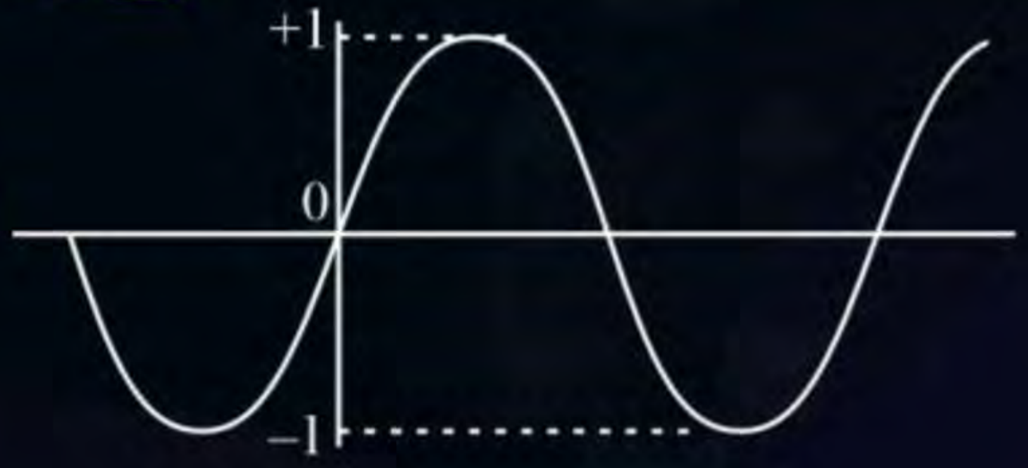
$f \to n$

$g \to n^{(1-1)} = n^0 = 1$

$f \geq_A g$

$n \geq n_0$

Sin x plot (wave)





Topic : Trichotomy Property

Observation from previous equation:

$f(n) = n$

$g(n) = n^{(1+\sin x)}$



- f(n) & g(n) are not holding the trichotomy property as they can't get a clear asymptotic comparison between both for $n \geq n_0$.



Topic : Trichotomy Property

Conclusion:

- Asymptotic notations may or may not follow Trichotomy Property (holds sometimes)
- Hence, property does not hold

  (As property holds if its always satisfied for all cases)

1) $f = O(g) \Longrightarrow f \leq_A g$

2) $f = \Omega(g) \Longrightarrow f \geq_A g$

3) $f = \Theta(g) \Longrightarrow f =_A g$

4) $f = o(g) \Longrightarrow f <_A g$

5) $f = \omega(g) \Longrightarrow f >_A g$



1) Given $f(n), g(n)$

and $f(n) = O(g(n))$

then

True?

D) None

Statement 1 : $f(n) = O((f(n)^2)) \Rightarrow$ False

Stat 2 : $2^{f(n)} = O(2^{g(n)}) \Rightarrow$ False

eg1: $f(n) = n$

$f(n)^2 = (n)^2 = n^2$

$n < n^2$

$f(n) < (f(n))^2$

(T)

$f(n) = O(f(n)^2)$

eg2 :- $f(n) = \frac{1}{n}$ or $\left(\frac{1}{\sqrt{n}} \text{ or } \frac{1}{\log n}\right)$

$f(n)^2 = \frac{1}{n^2}$

$\left(\frac{1}{n} > \frac{1}{n^2}\right) \longrightarrow f(n) > f(n)^2$

$f(n) \neq O(f(n)^2)$

$$2^{f(n)} = O(2^{g(n)})$$

eg:

$f(n) = n$
$g(n) = 1/n$

$f(n) = n$
$g(n) = \sqrt{n}$

invalid eg

eg2:

$f(n) = n$

$g(n) = n^2$

$f = O(g)$

$2^{f(n)} \quad 2^{g(n)}$

$2^n < 2^{n^2} \longrightarrow 2^{f(n)} = O(2^{g(n)})$

eg3:-

$f(n) = 3n$

$g(n) = n$

$3n \qquad n$

$2^{3n} \qquad 2^n$

$2^{(n+n+n)}$

2^n

$2^n \times 2^n \times 2^n$ ~~2^n~~

$2^n \times 2^n > 1$

$a^{x+y} = a^x \times a^y$

$2^{f(n)} \neq O(2^{g(n)})$



Summary

Asymptotic notations does not hold Tricotonomy property



THANK - YOU